

Euler-Cauchy Equation

①

$$x^2 y'' + x y' - y = x^3$$

$$\longrightarrow y'' = \frac{d^2 y}{dx^2}, \quad y' = \frac{dy}{dx}$$

$$\text{put } x = e^t$$

$$\frac{dx}{dt} = e^t$$

$$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx} = e^{-t} \frac{dy}{dt}$$

$$\begin{aligned} \frac{d^2 y}{dx^2} &= \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{dx} \left(e^{-t} \frac{dy}{dt} \right) = \frac{d}{dt} \left(e^{-t} \frac{dy}{dt} \right) \cdot \frac{dt}{dx} \\ &= -e^{-t} \cdot \frac{dy}{dt} \cdot e^{-t} + e^{-t} \cdot e^{-t} \cdot \frac{d^2 y}{dt^2} \\ &= e^{-2t} \left(\frac{d^2 y}{dt^2} - \frac{dy}{dt} \right) \end{aligned}$$

$$* \quad x^2 y'' + x y' - y = x^3$$

$$e^{2t} \left(e^{-2t} \left(\frac{d^2 y}{dt^2} - \frac{dy}{dt} \right) \right) + e^t \cdot e^{-t} \frac{dy}{dt} - y = e^{3t}$$

$$\Rightarrow \frac{d^2 y}{dt^2} - \frac{dy}{dt} + \frac{dy}{dt} - y = e^{3t}$$

$$\Rightarrow \frac{d^2 y}{dt^2} - y = e^{3t} \Rightarrow (D^2 - 1)y = e^{3t}$$

$$y_{\text{hom}} \rightarrow (D^2 - 1)y = 0 \quad \text{ch \&g} \quad x^2 - 1 = 0$$

$$x = \pm 1$$

$$y_{\text{hom}} = C_1 e^t + C_2 e^{-t} \Rightarrow C_1 x + \frac{C_2}{x}$$

$$y_{\text{non homo}} = \underline{\text{p.r.}} \quad \frac{1}{n^2 - 1} e^{3t} = \frac{1}{8} e^{3t}$$

$$y_{\text{homogeneous}} = \underline{\text{P.I.}} \quad \frac{1}{D^2-1} e^{-x} = \frac{1}{8} e^{3x}$$

Complete solution $y = C_1 x + \frac{C_2}{x} + \frac{1}{8} x^3$

• $x^2 y'' - x y' + 2y = x \log x$
 $\log x = t \quad \leftarrow x = e^t \Rightarrow \frac{dx}{dt} = e^t \Rightarrow \frac{dt}{dx} = e^{-t}$

$$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx} = e^{-t} \frac{dy}{dt}$$

$$\begin{aligned} \frac{d^2 y}{dx^2} &= \frac{d}{dx} \left(e^{-t} \frac{dy}{dt} \right) = \frac{d}{dt} \left(e^{-t} \frac{dy}{dt} \right) \cdot \frac{dt}{dx} \\ &= -e^{-t} \frac{dy}{dt} \cdot e^{-t} + e^{-t} \frac{d^2 y}{dt^2} e^{-t} = e^{-2t} \left(\frac{d^2 y}{dt^2} - \frac{dy}{dt} \right) \end{aligned}$$

$$e^{2t} \left(e^{-2t} \left(\frac{d^2 y}{dt^2} - \frac{dy}{dt} \right) \right) - e^t \cdot e^{-t} \frac{dy}{dt} + 2y = e^t \cdot t$$

$$\Rightarrow \frac{d^2 y}{dt^2} - 2 \frac{dy}{dt} + 2y = t \cdot e^t$$

y_{hom} $\frac{d^2 y}{dt^2} - 2 \frac{dy}{dt} + 2y = 0$

Ch Eq $x^2 - 2x + 2 = 0$
Roots $x = 1 \pm i$

$$y_{\text{hom}} = e^t (C_1 \sin t + C_2 \cos t)$$

• P.I. $\frac{1}{x^2 - 2x + 2} e^t \cdot t \Rightarrow e^t \frac{1}{(D+1)^2 - 2(D+1) + 2} \cdot t$

• P.I, $\frac{1}{D^2 - 2D + 2} e^t \cdot t \Rightarrow e^t \frac{1}{(D+1)^2 - 2(D+1) + 2} \cdot t$

$\Rightarrow e^t \frac{1}{D^2 + 2D + 1 - 2D - 2 + 2} \cdot t \Rightarrow e^t \frac{1}{(D^2 + 1)} \cdot t$

$\Rightarrow e^t (1 + D^2)^{-1} \cdot t = e^t (1 - D^2)t = t e^t$

Complete solution :- $y = e^t (C_1 \sin t + C_2 \cos t) + t e^t$

$y = x (C_1 \sin(\log x) + C_2 \cos(\log x)) + \log x \cdot x$

Bounded function :- A function $f(x, y)$ is said to be bounded if $|f(x, y)| \leq R$, for some fixed constant R , when $(x, y) \in R$, where R is some region in $x-y$ plane.

Example :- (1) $R = \{ (x, y) \mid \underbrace{|x| < 1}_{-1 < x < 1}, \underbrace{|y| < 1}_{-1 < y < 1} \}$

$f(x, y) = x^2 + y^2 \Rightarrow |f(x, y)| < 2 \quad \forall (x, y) \in R$

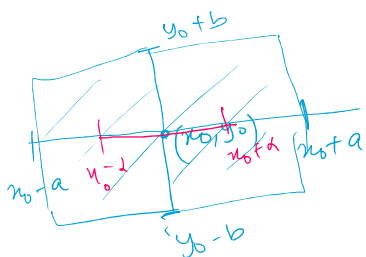
(2) $f(x, y) = \tan(x+y)$ if $|x+y| < \pi/2$
 $\hookrightarrow$ Not bounded $\because \tan \pi/2 \rightarrow \infty$.

Existence Theorem :- If $f(x, y)$ is continuous at all points in the rectangle R

$|x - x_0| < a$ and $|y - y_0| < b$

and bounded in R , i.e.

$|f(x, y)| < k$ for all $(x, y) \in R$



$\phi(x)$

$D_1, \dots, D_n$

$y_0 = b$

$$|f(x, y)| < K \text{ for all } (x, y)$$

$y = f(x)$

Then initial value problem $y' = f(x, y)$,

$y(x_0) = y_0$ has at least one solution. This solution is defined at least for all x , in the interval $|x - x_0| < \alpha$, where $\alpha = \min\{a, \frac{b}{K}\}$.

Example:- Consider the problem

$$\frac{dy}{dx} = 1 + y^2, \quad y(0) = 0$$

$$R = |x| < 5, \quad |y| < 3$$

Discuss the existence of solution of above I.V.P.

Solution:-

$$f(x, y) = 1 + y^2$$

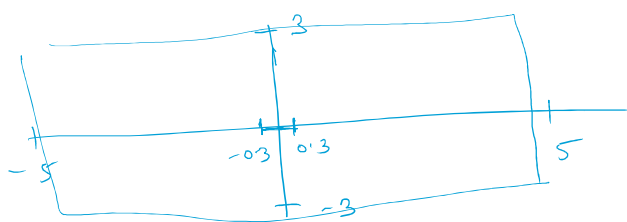
continuous function

$$|f(x, y)| = |1 + y^2| \leq 10 = K$$

• Solution exists

$$x \in I, \quad I \ni |x| < \alpha = 0.3$$

$$\alpha = \min\left\{5, \frac{3}{10}\right\} = \underline{0.3}$$



$$y = \tan x$$

Uniqueness Theorem:- If $f(x, y)$ and $\frac{\partial f}{\partial y}$ are continuous for all (x, y) in rectangle

$$R: |x - x_0| \leq a, \quad |y - y_0| \leq b$$

Also both $f(x, y)$ and $\frac{\partial f}{\partial y}$ are bounded in R

also $f(x, y)$

$\frac{\partial f}{\partial y}$

i.e. $|f(x, y)| \leq K$, $|\frac{\partial f}{\partial y}| \leq m$ $\forall (x, y) \in R$

Then the initial value problem $y' = f(x, y)$, $y(x_0) = y_0$ has at most one solution $y(x)$.

• Picard's Iteration Method

Consider IVP $\frac{dy}{dx} = f(x, y)$, $y(x_0) = y_0$

Solution is $\int dy = \int f(x, y) dx + C$

$\Rightarrow y = \int f(x, y) dx + C$

Particular solution of I.V.P

$y = y_0 + \int_{x_0}^x f(x, y) dx$

Does this satisfy initial condition $y(x_0) = y_0$

First iteration :-

$y_1 = y_0 + \int_{x_0}^x f(x, y_0) dx$

second iteration :-

$y_2 = y_0 + \int_{x_0}^x f(x, y_1) dx$

Third iteration :-

$y_3 = y_0 + \int_{x_0}^x f(x, y_2) dx$

Example :- Using Picard's approximation method, obtain the solution of the following diff Eqn upto five iterations, $\frac{dy}{dx} = y + x$, given $y(0) = 1$.

Solution:- $y = 1 + \int_0^x (y+x) dx$

(I) $y = 1 + \int_0^x (1+x) dx = 1 + \left(x + \frac{x^2}{2}\right) \Big|_0^x = 1 + x + \frac{x^2}{2}$

(II) $y = 1 + \int_0^x \left(1 + 2x + \frac{x^2}{2}\right) dx = 1 + x + x^2 + \frac{x^3}{6}$

(III) $y = 1 + x + x^2 + \frac{x^3}{3} + \frac{x^4}{24}$

(1) $\frac{dy}{dx} = y + x$, $y(0) = 1$

$\Rightarrow \frac{dy}{dx} - y = x$, $y(0) = 1$

I.F. e^{-x}

Solution:- $y \times e^{-x} = \int x \cdot e^{-x} dx + C$

$y e^{-x} = x \frac{e^{-x}}{-1} - \int \frac{e^{-x}}{-1} dx + C$

$y e^{-x} = -x e^{-x} - e^{-x} + C$

$y = -x - 1 + C e^x$, $y(0) = 1$

$1 = -0 - 1 + C \Rightarrow \underline{C = 2}$

Particular solution

$y = -x - 1 + 2e^x$

$x^2 \quad x^3 \quad x^4 + \frac{x^5}{5} + \dots$

1. sum

u

$$2e^x = 2 \left(1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \dots \right)$$

$$2e^x - x - 1 = 1 + x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{12} + \dots$$

Convergence of Picard's iteration method

Under the condition of Existence and Uniqueness thm, the seq of functions in Picard's iteration method, converges to solution of I.V.P.